Machine Learning Prediction Project

IFJC4002 | MC Practical Artificial Intelligence with Python

Faculty of Intelligent Computing
Universiti Malaysia Perlis
Course coordinator: Ir. Dr. Marni Azira Markom

Project question

Working in a group of five, develop a machine learning prediction system for one lecturer-approved case. Analyse the problem and dataset, design and compare at least two suitable models, and integrate the selected model into an interactive Streamlit dashboard. Test and deploy the application, then demonstrate and explain your solution.

The class will form six groups. Each group must select a different case from the ten options on the next page. This project uses tabular data for regression or classification. Image classification and object detection are outside the project scope.

Approved project case options

Choose one case and obtain lecturer approval before development. The lecturer will provide or approve the dataset. Confirm the available columns, prediction target and data source before finalising the scope. The inputs below are examples and may be adapted to the approved dataset.

No	Case	Example input features	Target and task
1	House price prediction	Floor area, bedrooms, bathrooms, property age, location	House price Regression
2	Used car price prediction	Vehicle age, mileage, engine size, fuel type, transmission	Resale price Regression
3	Bicycle rental demand prediction	Temperature, humidity, weather, hour, working-day status	Number of rentals Regression
4	Building energy consumption prediction	Floor area, occupancy, operating hours, outdoor temperature	Daily energy use Regression
5	Delivery time prediction	Distance, traffic, weather, delivery type, preparation time	Delivery duration Regression
6	Product quality prediction	Processing temperature, pressure, machine speed, material measurements	Good or Defective Classification

No	Case	Example input features	Target and task
7	Machine failure prediction	Temperature, rotational speed, torque, tool wear	Failure or No Failure Classification
8	Customer churn prediction	Subscription duration, service usage, monthly charges, contract type	Leave or Stay Classification
9	Hotel booking cancellation prediction	Booking lead time, stay duration, previous cancellations, deposit type	Cancelled or Not Cancelled Classification
10	Employee absenteeism prediction	Previous absences, workload, commuting distance, working hours	Hours absent Regression

Scope and data suitability

Use only features available when the prediction would be made. Exclude identifiers and fields that reveal the outcome. For bicycle demand, exclude component counts that sum to the target. For booking cancellation, exclude final reservation status and post-cancellation information.

Case 4 must use a different approved dataset and scope from the household electricity example demonstrated in class. Use public or lecturer-provided data; cite the source and follow its permitted use. Do not include identifiable employee or customer records in the dashboard.

Required project tasks

Task 1 Analyse the problem and data CO1

Define the intended user, practical problem and benefit of the prediction. Identify the target, units or class labels, input features and task type. Analyse the dataset size, feature types, missing values, unusual values, distributions and class balance where relevant. Explain suitable AI approaches and the implications of the data characteristics for model performance.

Task 2 Design develop and evaluate models CO2

Design a clear workflow from data preparation to prediction. Justify feature selection, missing-value treatment, categorical encoding and scaling where needed. Fit learned preprocessing on the training data only. Use an appropriate training and testing split; preserve chronological order for forecasting and prevent the same entity appearing in both sets when relevant.

Train and compare at least two different suitable algorithms on the same evaluation split. Examples include linear regression and random forest regression, or logistic regression and decision tree classification. Explain the model settings and include a simple baseline, such as predicting the training mean or majority class. Use validation or cross-validation for model selection where feasible; reserve the test set for final evaluation.

For regression, report MAE, RMSE and R-squared. For classification, report a confusion matrix, accuracy, precision, recall and F1-score, stating the positive class and averaging method where applicable. Discuss errors and any class imbalance. Select and justify the final model using performance and practical constraints such as speed, interpretability and available resources. Explain limitations and realistic improvements.

Task 3 Implement test deploy and monitor CO3

Use Python to implement a reproducible workflow. Save and reload the model with its preprocessing so the dashboard uses the same transformations and feature order. Build a Streamlit dashboard with a clear title, brief instructions, suitable input controls, input validation and clearly labelled predictions with units or class names.

Deploy the application through GitHub and Streamlit Community Cloud. Provide the app link and test it against notebook predictions. Check typical inputs, boundary values and invalid inputs. Demonstrate basic monitoring using application logs or non-sensitive records of inputs and predictions, and identify one sign that the model may need review. New predictions alone do not establish real-world accuracy without true outcomes.

AI assisted development

You may use ChatGPT for coding support. Retain at least five representative prompts covering different development stages. Explain how the generated code was checked, corrected or adapted. All members must understand the submitted work and be able to explain their contributions.

Submission and demonstration

Group submission

- Report: one PDF per group, approximately 8 to 12 pages excluding the cover, references and appendices.
- Code: a runnable Google Colab notebook or Python training script, app.py, requirements.txt, saved model and preprocessing files, and a README with reproduction and deployment instructions.
- Data: include the approved dataset where redistribution is permitted. Otherwise provide the source link and clear download instructions. State any filters or sampling used.
- Links: provide the GitHub repository and working Streamlit application links. Give the lecturer access where required.
- Evidence: include a comparison of model results, dashboard screenshots, a test log with inputs, expected behaviour and actual results, and a short record of monitoring observations.
- Contribution and prompt records: list all five members with student IDs, specific tasks and supporting evidence; include at least five prompts and explanations of verification or modification.

Suggested report structure

1. Problem definition and dataset analysis (CO1).
2. Solution design and preprocessing (CO2).
3. Model development and comparison (CO2).
4. Final model selection, limitations and improvements (CO2).
5. Implementation, dashboard, deployment, testing and monitoring evidence (CO3).
6. Conclusion and references.

Appendices: contribution record, prompt record and additional test evidence.

Demonstration format

Suggested duration: 10 minutes for the group demonstration and 5 minutes for questions. All five members must participate. Show a live prediction, explain the model comparison, test one invalid or boundary input, show monitoring evidence and answer questions about the code. The lecturer will announce the submission deadline and demonstration schedule.

How the rubric is scored

Each criterion has four performance levels: Excellent = 4, Good = 3, Satisfactory = 2 and Limited = 1. Award 0 when there is no assessable evidence or the criterion is not attempted. Select the level that best matches the submitted evidence.

Criterion mark = (level awarded / 4) x criterion maximum. For example, Good on a 10-mark criterion earns 7.5 marks. Sum all criteria for a final mark out of 80; retain quarter marks. Do not apply an additional 80% multiplier to the raw total.

The report and implementation are group assessment items. Every member must document their contribution and demonstrate understanding. Record individual participation during the demonstration; any individual mark adjustment must follow the lecturer's announced assessment policy.

Report rubric CO1 and CO2 40 marks

Assess the analysis and justification in the report. Coding execution and live operation are assessed separately under CO3. Apply the scoring rule on the preceding page.

Criterion	Excellent 4	Good 3	Satisfactory 2	Limited 1
Problem and approach CO1 5 marks	Clear user, target, task and inputs; analyses why suitable AI approaches address the case.	Clear problem and target; suitable approach with some analytical justification.	Basic problem and target; approach stated with limited explanation.	Problem or target unclear; approach weakly matched or unsupported.
Data and performance analysis CO1 5 marks	Analyses data characteristics and model behaviour; connects patterns, imbalance or errors to practical consequences.	Relevant data exploration and performance interpretation; minor gaps in connections.	Basic summaries and results; limited interpretation of their implications.	Minimal exploration; results largely listed, misinterpreted or unsupported.
Solution design CO2 5 marks	Coherent workflow; justified features, preprocessing and evaluation design; leakage risks addressed.	Suitable workflow and preprocessing; evaluation mostly sound with minor omissions.	Workable workflow; weak justification or incomplete treatment of data risks.	Incomplete or unsuitable design; major preprocessing or evaluation flaws.
Model development CO2 10 marks	Two suitable algorithms and baseline developed; settings and training decisions clearly explained and reproducible.	Two suitable algorithms and baseline developed; most decisions explained.	Two models attempted; baseline, settings or development rationale incomplete.	Only one usable model or major development gaps; little defensible rationale.
Evaluation and model selection CO2 10 marks	Fair comparison; correct metrics and test separation; errors analysed; final selection supported by evidence.	Valid comparison and mostly correct metrics; reasonable selection with minor analytical gaps.	Some valid results; comparison, metric choice or selection justification is weak.	Comparison unreliable or absent; inappropriate metrics, leakage or unsupported selection.
Constraints and improvements CO2 5 marks	Weighs practical constraints and limitations; proposes specific feasible improvements linked to findings.	Relevant constraints and limitations; useful improvements with some links to evidence.	General limitations and improvements; feasibility or links to findings underdeveloped.	Constraints largely ignored; limitations or improvements vague or unrealistic.

Coding skills rubric CO3 20 marks

Assess the submitted code and supporting execution evidence. Each criterion carries 5 marks. A polished report alone does not establish implementation competence.

Criterion	Excellent 4	Good 3	Satisfactory 2	Limited 1
Reproducible Python workflow 5 marks	Code runs end to end; paths, dependencies and random seeds handled; clear reusable structure and instructions.	Workflow runs with minor setup changes; dependencies and instructions mostly complete.	Workflow partly reproducible; manual fixes or unclear dependencies needed.	Major execution failures; missing files or unclear workflow prevent reproduction.
Model and dashboard integration 5 marks	Saved model and preprocessing reload correctly; inputs align with training; dashboard and notebook agree on checked cases.	Integration works; minor inconsistencies or limited agreement checks.	Basic integration works for some inputs; preprocessing or feature handling needs correction.	Integration largely fails; incorrect input mapping or preprocessing makes predictions unreliable.
Input validation and testing 5 marks	Appropriate controls and validation; documented normal, boundary and invalid-input tests; failures handled clearly.	Suitable controls and several relevant tests; minor validation or handling gaps.	Basic controls and limited tests; boundary or invalid-input checks incomplete.	Minimal validation or testing; common inputs cause avoidable failures.
Deployment and monitoring 5 marks	Accessible deployed app; complete deployment files; useful logs or records inspected and a review trigger identified.	App deployed; setup mostly documented; basic monitoring evidence with minor gaps.	Deployment partly works or monitoring evidence is minimal; setup requires assistance.	Deployment largely unsuccessful; monitoring absent or unusable.

Evidence the assessor may inspect

Training notebook or script; saved model and transformations; app.py; requirements.txt; README; GitHub repository; deployed application; notebook-to-dashboard prediction checks; test log; application logs or monitoring record.

Demonstration rubric CO3 20 marks

Assess practical operation and explanations during the demonstration. Each criterion carries 5 marks. All five members must participate and be available for questions.

Criterion	Excellent 4	Good 3	Satisfactory 2	Limited 1
Live application operation 5 marks	Independently demonstrates complete input-to-prediction workflow; clearly interprets output and checks a new valid case.	Demonstrates working workflow and interprets output with minor prompting.	Demonstrates part of workflow; output interpretation or operation needs assistance.	Cannot reliably operate the application or explain the displayed prediction.
Code understanding and contribution 5 marks	All members explain their own code and contributions; trace preprocessing and prediction flow; justify AI-generated code checks.	Most explanations accurate; contributions supported; minor gaps across members.	Uneven participation or basic explanations; contribution or verification evidence limited.	Members cannot explain key code or contributions; little evidence of checking generated code.
Testing and troubleshooting 5 marks	Demonstrates boundary or invalid input; explains expected behaviour; locates relevant code and reasons through a fault.	Demonstrates a useful test and explains handling; troubleshooting mostly sound.	Limited test demonstration; identifies an issue but needs help to diagnose it.	Testing absent or ineffective; unable to explain errors or troubleshooting steps.
Deployment and monitoring practice 5 marks	Shows live deployment and monitoring evidence; explains a review trigger, limitations and how to update the app.	Shows deployment and basic monitoring; explains most maintenance steps.	Partial deployment or monitoring demonstration; limited explanation of maintenance.	Unable to show usable deployment or monitoring, or explain how the app is maintained.